

SIT225 Data Capture Technologies

Pass Task: Connecting the sensor with Arduino



```
sketch_mar6a_copy_20250320135436.ino
1 // Simple LDR reading program for Arduino Nano 33 IoT
2 const int ldrPin = A0; // LDR connected to analog pin A0
3 unsigned long previousMillis = 0;
4 const long interval = 5000; // Data collection interval (5 seconds)
5
6 void setup() {
7   Serial.begin(9600); // Initialize serial communication
8   while (!Serial); // Wait for Serial to be ready
9
10  // Print CSV header
11  Serial.println("Time(ms),Light_Level");
12 }
13
14 void loop() {
15   unsigned long currentMillis = millis();
16
17   // Read sensor at defined intervals
18   if (currentMillis - previousMillis >= interval) {
19     previousMillis = currentMillis;
20
21     // Read the analog value from the LDR
22     int lightLevel = analogRead(ldrPin);
23
24     // Print the data to the serial monitor
25     Serial.print(currentMillis);
26     Serial.print(",");
27     Serial.println(lightLevel);
28   }
29 }
```

Output Serial Monitor x

Message (Enter to send message to 'Arduino NANO 33 IoT' on 'COM3')

7225000,1023
7230000,1022
7235000,1023
7240000,1023
7245000,1023
7250000,1021
7255000,1023
7260000,1023
7265000,1023
7270000,1023
7275000,1023

Updates are available for some of your libraries.

LATER INSTALL MANUALLY INSTALL ALL

Ln 15, Col 42: Arduino NANO 33 IoT on COM3 1

summary

The Arduino code reads the light data from the LDR sensor in every 5 seconds. It sends the light values and the time to the serial monitor in CSV format, and the data format is time in milliseconds followed by the light sensor value. The data can be used for logging and analysis, and this system is ideal for the smart lighting monitoring applications, like smart lighting as well as environmental sensing.

```
python.py X ldr_readings.csv vasu.py ldr.py ldr_data ccc.csv
python.py > ...
1 import serial
2 import time
3 import csv
4
5 # ===== CONFIGURATION =====
6 PORT = 'COM3'
7 BAUD_RATE = 9600
8 CSV_FILENAME = 'ldr_readings.csv'
9 LOG_DURATION = 3600
10
11 # ===== CONNECT TO ARDUINO =====
12 try:
13     arduino = serial.Serial(PORT, BAUD_RATE, timeout=1)
14     time.sleep(2)
15     print(f"[INFO] Connected to Arduino on {PORT}")
16 except Exception as e:
17     print(f"[ERROR] connection error: {e}")
18     exit()
19
20 # ===== OPEN CSV FILE =====
21 with open(CSV_FILENAME, mode='w', newline='', encoding='utf-8') as file:
22     writer = csv.writer(file)
23     writer.writerow(["timestamp", "Milliseconds", "LightValue"])
24
25 print(f"[INFO] Logging data to '{CSV_FILENAME}' for {LOG_DURATION} seconds...")
26 start_time = time.time()
27
28 while time.time() - start_time < LOG_DURATION:
29     try:
```

2025-05-15 13:19:25, 8460000, 1023
2025-05-15 13:19:30, 8445000, 1023
2025-05-15 13:19:35, 8450000, 1023
2025-05-15 13:19:40, 8455000, 1023
2025-05-15 13:19:45, 8460000, 1023
2025-05-15 13:19:50, 8465000, 1023

Summary:

In this code, I am shown to read serial data and connect to Arduino nano 33 IoT with COM3 and which collect the data into a CSV file, and my CSV file name is ldr_reading.csv. In this code, I can use the serial, csv, and time modules. After all the connections to the Arduino, the script enters a loop that continuously reads the data lines, which are printed by the Arduino.

